Heimdall: Discovering IPv6 Fully-Responsive Prefixes at Scale via Dual-Scan Hierarchical State-Space Models with In-Range-Aware Feedback

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***Abstract*—Fully-Responsive Prefixes (FRPs)—IPv6 prefixes whose every address replies to probes—are pivotal for Internet-wide measurement and target generation, yet the 128-bit address space renders exhaustive discovery infeasible. Existing generators such as *Luori* and *Gungnir* either exhaust the probing budget within a handful of Autonomous Systems (ASes), or waste a large fraction of it on candidates *already covered by the known seed set*. Our study shows that 32%–43% of the FRPs confirmed by state-of-the-art generators are strict sub-prefixes of the aligned public seed set, so the corresponding probes yield no fresh information. We present Heimdall, an online FRP-discovery framework built on a *Dual-Scan Hierarchical Mamba* (DS-HMamba) generator that jointly performs an *outer scan* over the 128-bit bit hierarchy and an *inner scan* across same-AS samples, fused by an AS-conditioned gate. Candidates are validated by a ping-style probing module, and Heimdall pre-fixes the known-FRP range so that probes only target *net-new* candidates; the resulting labels continually fine-tune DS-HMamba. Under identical budgets of 2–10 M packets, Heimdall discovers 2.8× more raw and 3.23× more net-new FRPs than Gungnir, covering 5.9× more address space and 3.2× more ASes while sampling 1.7× faster.**

***Index Terms*—IPv6 measurement, Fully-Responsive Prefix, Address-space discovery, Hierarchical state-space model, Mamba, Active probing, Online learning.**

I. Introduction

Internet-wide active scanning underpins modern Internet measurement [1]–[3], [24], [27]. While IPv4 has been mapped in its entirety [1], [36], the 2^{128} address space of IPv6 renders brute-force scanning infeasible and has motivated a rich family of Target-Generation Algorithms (TGAs) [4]–[8], [28]–[33], [35]. These TGAs are, however, undermined by a peculiar phenomenon: a non-negligible fraction of the addressable space consists of *Fully-Responsive Prefixes* (FRPs)—routing prefixes for which *every* address responds to probes [9]–[12], [34], [43]. FRPs distort measurement conclusions (hitlist inflation, service counts, reachability), waste bandwidth and complicate downstream analyses [14]. Discovering them precisely is therefore a first-class problem for the community.

Two recent works represent the state of the art. *Luori* [12] casts FRP discovery as a reinforcement-learning problem and expands a prefix tree via Monte-Carlo Tree Search. *Gungnir* [13] introduces a Transformer autoregressive model that unifies multiple protocols and applies a granularity-correction post-process. Both methods, however, share three limitations that we systematically diagnose in this paper. First, neither method excludes candidates that fall *inside* an already-known FRP range at generation time (**L1 – redundancy w.r.t. the known range**): in our large-scale reproduction, under a 10M probing budget 43% of Gungnir's confirmed FRPs are strict sub-prefixes of the aligned public seed set and 32% of Luori's are, so the corresponding probes contribute no fresh signal because a sub-prefix of an already-known FRP is fully responsive by definition (§V). Second, Transformer decoders treat each prefix in isolation (**L2 – weak intra-AS coupling**), yet FRPs from a single AS heavily share nibble-level patterns (§III); ignoring those correlations forces the model to re-discover the same substructures many times. Third, both baselines train once and freeze (**L3 – no online feedback**), while real deployments observe non-stationary signals (churn, new allocations) that cause un-adapted generators to drift.

Our approach: Heimdall. We propose **Heimdall**, an online FRP-discovery framework built around three ideas. *(i)* A **Dual-Scan Hierarchical Mamba** (DS-HMamba) generator: an *outer scan* along the 128-bit sequence captures the coarse→medium→fine bit hierarchy of an IPv6 prefix, while an *inner scan*, executed over same-AS samples at each bit position, mines the shared nibble structures particular to that AS. An AS-conditioned gate fuses the two scan outputs (§IV-C). *(ii)* A three-stage masked-language-model (MLM) decoder generates candidates coarse-to-fine, with temperature-controlled top-p sampling (§IV-D). *(iii)* An **in-range-aware feedback** loop (§IV-E) pre-fixes the known-FRP set at start-up and thereafter probes only candidates *outside* that set, turning every probe into net-new information; confirmed/rejected labels then fine-tune DS-HMamba on the fly.

Contributions. The primary contributions of this work are as follows:

- We are the first, to the best of our knowledge, to quantify the *in-range redundancy* of state-of-the-art FRP generators and to show that Gungnir spends 43% of its confirmed hits (and Luori 32%) on candidates strictly inside the aligned seed range (§V).
- We propose DS-HMamba, a dual-scan hierarchical state-space model that jointly exploits the intra-prefix bit hierarchy and the intra-AS sample correlations at $O(L)$ time per layer, strictly cheaper than the $O(L^2)$ self-attention used by Gungnir (§IV-C).
- We design a two-level SeedFilter and an in-range-aware online fine-tuning loop that turns wasted probes into productive discovery (§IV-E).
- Under identical 2 M–10 M probing budgets on the public

IPv6 routing view, Heimdall outperforms Gungnir by up to **2.8**× on raw confirmed FRPs and **3.23**× on net-new confirmed FRPs; **5.9**× on covered address space; **3.2**× on ASes and **1.76**× on countries. Against Luori the net-new gap is **1.35**× at 10M and grows monotonically with the budget (§VI-B).

II. Background and Related Work

A. Fully-Responsive Prefixes

Murdock *et al.* [16] first noticed the misleading effect of *aliased prefixes*—contiguous IPv6 ranges whose addresses reply uniformly to probes—on address inference. Zirngibl *et al.* [10] refined the definition and adopted the term FRP, formally requiring a 100% probe-reply rate across the prefix. Sattler *et al.* [11] report similar aliasing-driven inflation. Their operational relevance is now well-established: FRPs distort hitlists [9], [15], [43], inflate service counts and complicate downstream analyses [14], [26].

B. FRP Probing Methods

1) Aliased-Prefix Detection (APD) and MAPD: The seminal APD of Gasser *et al.* [9] probes 16 addresses, one per nibble, drawn from the sixteen child `0`–`f` subnets of a candidate prefix; if all 16 reply the prefix is declared aliased. Its exhaustive variant MAPD [10] enumerates all parent prefixes down to ℓ=64, but its cost is exponential in the announced prefix length and thus infeasible under a bounded probing budget.

2) Luori [12]: Luori casts FRP discovery as a Monte-Carlo Tree Search (MCTS) [41] problem and expands a prefix tree bit-by-bit. Its search depth grows linearly with prefix length, MCTS concentrates rollouts on seed-rich sub-trees (visible as the saturation of Luori's curve in Fig. 5(a)), and trees for different routing prefixes do not share statistics, so intra-AS patterns must be re-learned from scratch.

3) Gungnir [13]: Gungnir is a Transformer autoregressive generator [38], [39] that tokenises the prefix bit-by-bit and appends context tokens (BGP prefix, AS, organisation, business category, protocol). A causal cross-entropy loss is minimised on top of Top-p nucleus sampling [20] (best p=0.6), followed by a *granularity correction* step that re-probes each hit under alternative prefix lengths. We identify three structural weaknesses that motivate this paper: (i) 43% of confirmed FRPs at 10M are in-range w.r.t. the aligned public seed set (§V); (ii) $O(L^2)$ attention makes Gungnir $1.7-2.1\times$ slower per candidate than a linear scanner (§VI-D); (iii) the model is frozen after pre-training, so the redundancy ratio stays at $\sim$43% across the entire budget grid.

C. Selective State-Space Models

Mamba [17], [40] is a selective State-Space Model that reaches Transformer-level accuracy at *linear* time and memory in the sequence length. Its recurrence $h_t = \exp(\Delta_t A)\, h_{t-1} + \Delta_t B_t u_t$, $y_t = C_t^\top h_t$ uses a diagonal A and input-dependent gates (Δ_t, B_t, C_t), allowing the sequence to be evaluated by a parallel prefix scan in $O(L \log L)$ work. We argue in §IV-B that Mamba's linear cost and selective recurrence match FRP generation.

III. Problem Analysis and Motivation

A. Definitions

We work with three sets throughout. Let $\mathcal{S} \subset \{0,1\}^{\leq 128} \times \mathbb{N}$ denote the current *seed FRP set* (each entry is a prefix and its routing length); let $\mathcal{S}_0 \supseteq \mathcal{S}$ denote the *aligned public seed set* obtained by unioning $\mathcal{S}$ with third-party FRP releases and the APD-derived aggregations described in §VI-A. Given a probing budget B (in packets), $\mathcal{C}_\theta$ is the set of candidate prefixes generated by algorithm θ and $\mathcal{F}_\theta \subseteq \mathcal{C}_\theta$ the subset whose APD test succeeds. Because a probe that lands *inside* an already-known FRP $s \in \mathcal{S}_0$ can only re-confirm what is known, we measure discovery quality by the *net-new coverage*

$$\mathrm{NN}(\theta) \;=\; \big|\{c \in \mathcal{F}_\theta \;:\; \nexists\, s \in \mathcal{S}_0,\; c \preceq s\}\big|, \tag{1}$$

where $c \preceq s$ means c is a (non-strict) sub-prefix of s. The corresponding *redundancy ratio* of θ is $\rho(\theta) = 1 - \mathrm{NN}(\theta)/|\mathcal{F}_\theta|$; ρ close to zero means every probe contributes new information, while ρ close to one means the generator merely re-discovers what the community has already published. Section V shows that state-of-the-art generators sit at $\rho \approx 0.32 - 0.43$.

B. In-Range Probes Waste a Large Share of the Budget

Under a 10M-packet budget Gungnir confirms 1,675 FRPs, but a full sub-prefix check against $\mathcal{S}_0$ (via Eq. (1)) reveals that only 954 are genuinely *net-new*; the remaining 721 (43.0%) fall inside a known FRP. Luori exhibits a milder but still material redundancy of 31.9% (Fig. 3, dissected in §V). Two observations drive our design: (a) probing a sub-prefix of a known FRP is *information-free*: every $c \preceq s$ is fully responsive by definition; and (b) $\mathcal{S}_0$ is available *before* probing starts, so a $O(\log |\mathcal{S}_0|)$ trie lookup can refuse redundant candidates. Heimdall builds this lookup into a two-level SeedFilter (§IV-E) that guarantees every issued probe lands outside $\mathcal{S}_0$.

C. FRPs Are Strongly Regular Within Each AS

Following [13], we quantify the regularity of the FRP pattern by $H_{\text{pattern}}(\mathcal{S}) = -\sum_x P(x) \log P(x)$, where $P(x)$ is the empirical frequency of the first-ℓ-bit pattern x; the maximum is attained by a uniform distribution. Fig. 1 plots H_{pattern} for the seed set aggregated *globally* and partitioned by business category, sub-category, organisation and AS. All curves sit far below the theoretical upper bound; the gap widens monotonically as we condition on finer partitions, and for ℓ=20 the entropy drops from ~ 12.5 (global) to ~ 5.2 (top-1 AS), a 2.4× compression. In short, *conditioning on the AS collapses the pattern space*, yet the intra-AS residual is still non-trivial (the model must exploit AS conditioning but not collapse to a single per-AS template).

Two consequences follow. First, a generator that models each prefix in isolation (as Gungnir does) must re-derive intra-AS substructure from scratch, spending $O(L^2)$ attention on an axis already exposed by the AS label. Second, the intra-AS

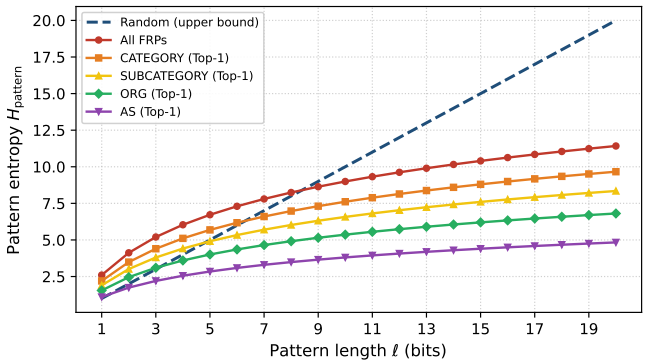


Fig. 1: Pattern entropy $H_{\text{pattern}}(\mathcal{S})$ of the seed FRP set as a function of the pattern length ℓ. The dashed line marks the theoretical upper bound of a uniformly random distribution. Partitioning the seed set by business category, sub-category, organisation and AS pushes the observed entropy progressively below the bound, revealing strong *intra-AS* regularity that Heimdall exploits through an inner Mamba scan (§IV-C).

regularity motivates a *second* scan axis that runs across same-AS samples, learning the shared nibble structures once. This is the role of the *inner scan* in §IV-C.

D. Objective

Combining the two observations, we formalise Heimdall's design objective as

$$\max_{\theta} \ \mathrm{NN}(\theta) \ \text{ s.t. } \ |\mathcal{C}_\theta| \le B/16, \ \ c \not\preceq s \ \forall c \in \mathcal{C}_\theta, \ s \in \mathcal{S}_0, \quad (2)$$

i.e. maximise net-new coverage subject to a probing budget and the in-range exclusion constraint. The next section describes an architecture that fulfils both constraints efficiently.

IV. Design of Heimdall

A. Overview

Fig. 2 shows the pipeline of Heimdall. The **Seed Store** holds $\mathcal{S}_0$ and metadata (AS/prefix), loaded once at start-up. The **DS-HMamba generator** combines an *outer* scan along the 128-bit axis with an *inner* scan along the same-AS axis, fused by an AS-conditioned gate. A three-stage **MLM decoder** emits a candidate c coarse-to-fine, and a two-level **SeedFilter** removes duplicates and in-range candidates w.r.t. $\mathcal{S}_0$ *before* probing. The **Active Prober** issues 16 APD probes per surviving candidate with a bounded retry budget; confirmed FRPs enter $\mathcal{F}$ and, together with negatives, refresh DS-HMamba every $\{0.5, 2.5, 4.5, 6.5\}$ M probes via **online fine-tuning**. Compared to Luori (open-loop MCTS) and Gungnir (open-loop autoregression), this is the first FRP architecture whose generator is *reactively adapted* to observed outcomes.

B. Why a Selective State-Space Model?

A good FRP generator must satisfy four requirements: (R1) modelling long-range bit dependencies along the length-L=128 prefix [18], [42]; (R2) linear cost at inference—a 10M budget corresponds to $\sim$ 625K generations, and an $O(L^2)$ attention cost incurs the main bottleneck of Gungnir on a single RTX 4090 (Table IV(b)); (R3) input-selective memory, since the effective alphabet varies sharply by AS (Fig. 1); and (R4) composability along a second (same-AS) axis without quadratic blow-up. LSTM/GRU are linear (R2) but lack input-dependent gating (fail R3); Transformer decoders address R1 and R3 through attention but are $O(L^2)$ (fail R2) and not naturally composable along a second axis (fail R4). Selective SSMs (Mamba [17], [40]) uniquely satisfy all four: their recurrence $h_t = \exp(\Delta_t A) h_{t-1} + \Delta_t B_t u_t$, $y_t = C_t^\top h_t$ is $O(L)$ per layer (R2), the input-dependent gates (Δ_t, B_t, C_t) realise selective memory (R3), and the diagonal A makes the parallel scan axis-agnostic—composable along an AS axis (R4).

C. Dual-Scan Hierarchical Mamba (DS-HMamba)

Bit-hierarchy embedding. A prefix $b_{1:L} \in \{0,1\}^L$ is split into three semantic layers [18]: coarse $\mathcal{L}_c = [1, \ell_c]$ (routing + RIR, ℓ_c=32), medium $\mathcal{L}_m = [\ell_c{+}1, \ell_m]$ (subnet, $\ell_m \approx$ 48–64) and fine $\mathcal{L}_f = [\ell_m{+}1, L]$ (interface identifier). Each bit is lifted to $x_i \in \mathbb{R}^d$ with a learned embedding plus a layer-specific offset $e_s \in \{e_c, e_m, e_f\}$.

Outer scan (bit axis). N_o Mamba blocks operate on $x_{1:L}$, interleaved with a gated Conv-1D of width 4 and a residual FFN, producing $y^{\text{outer}} \in \mathbb{R}^{L \times d}$. The scan captures the coarse$\rightarrow$medium$\rightarrow$fine hierarchy at $O(L)$ per block (R1, R2), while the input-dependent Δ_t realises R3.

Inner scan (same-AS axis). For each bit position ℓ we gather the ℓ-th feature of all same-AS prefixes in the current batch $\mathcal{B}_\mathcal{A}$, forming a $|\mathcal{B}_\mathcal{A}| \times d$ sequence per AS. A second stack of N_i Mamba blocks scans this axis, producing y_ℓ^{inner}. This captures shared nibble structures across same-AS prefixes without the $O((|\mathcal{B}_\mathcal{A}|L)^2)$ cost of a joint attention grid, and requires no code change (R4).

AS-conditioned gating fusion. An AS-embedding $e_\mathcal{A} \in \mathbb{R}^d$ conditions a soft gate that fuses the two scans position-wise:

$$g_\ell = \mathrm{softmax}(W_g [e_\mathcal{A}; y_\ell^{\text{outer}}; y_\ell^{\text{inner}}]), \quad z_\ell = g_{\ell,1} y_\ell^{\text{outer}} + g_{\ell,2} y_\ell^{\text{inner}}. \quad (3)$$

The gate favours the outer scan for well-populated ASes and the inner scan for ASes with rich intra-set regularity (the low-entropy tail of Fig. 1).

Three-stage MLM training. We train DS-HMamba with a three-stage masked-language-model loss [19] restricted to the coarse/medium/fine bit groups: at each step one of $\{c, m, f\}$ is chosen uniformly, its bits are masked, and the model reconstructs them from the unmasked bits and context tokens. The total loss is $\mathcal{L}_{\text{MLM}} = -\sum_{s \in \{c,m,f\}} \mathbb{E}_{\mathcal{L}_s} \sum_{i \in \mathcal{L}_s} \log P_\theta(b_i | b_{\setminus \mathcal{L}_s}, \text{ctx})$. Unlike Gungnir's causal cross-entropy, the MLM formulation is *bidirectional*: fine bits can attend to coarse bits and vice versa, matching the non-causal nature of FRP masks.

D. Three-Stage Coarse-to-Fine Inference

At inference we invert the training curriculum. Given the target routing prefix and its context (AS, organisation, category, protocol), the decoder generates a candidate in three passes using Top-p nucleus sampling [20] with τ=1: (1) sample coarse bits with p=0.6; (2) fix them and sample medium bits

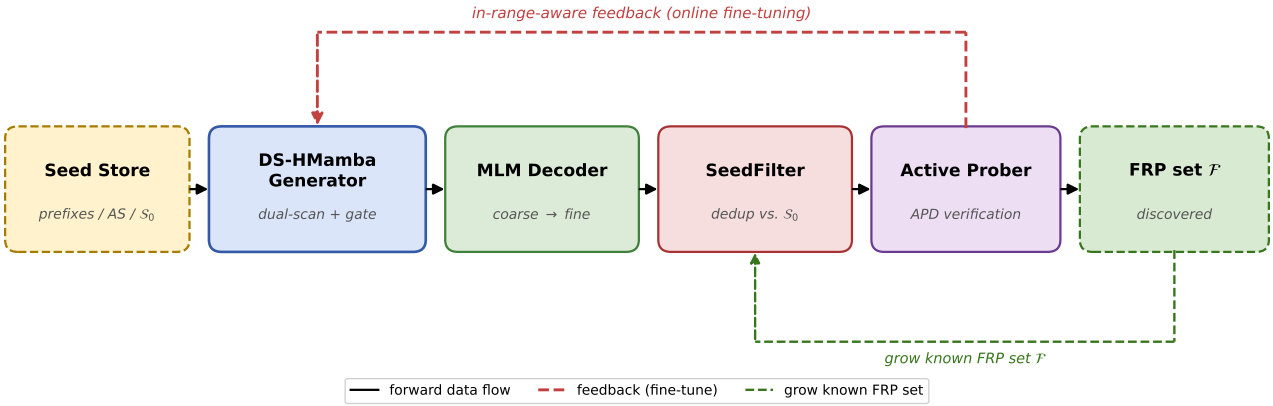


Fig. 2: Architecture of Heimdall. Let $\mathcal{S}_0$ denote the public seed FRP set loaded at start-up (§III-A). Solid arrows show the forward flow (generation→decoding→filter→probing); the red dashed path realises the *in-range-aware feedback* loop that distinguishes Heimdall from Luori and Gungnir.

with p=0.6; (3) fix coarse+medium and sample fine bits with p_f=0.4 (a tighter nucleus avoiding explosive branching). The choice p=0.6 coincides with the optimum reported by [13].

E. Two-Level SeedFilter and In-Range-Aware Feedback

Data structure. $\mathcal{S}_0$ is loaded into a Patricia trie keyed by the routing prefix; each node stores the routing length, so the sub-prefix test $c \prec s$ is answered in $O(\ell(c))$ time ($\sim$2 μs per lookup on a single core, four orders of magnitude cheaper than an APD round-trip).

Filter chain. For every candidate c: (L1) if $\exists s \in \mathcal{F} : c{=}s$, drop silently (no negative); (L2) if $\exists s \in \mathcal{F} : c \prec s$, drop silently (would only re-confirm known coverage); (L3) otherwise dispatch to the Active Prober. Rejected candidates are gradient-free and never enter fine-tuning as artificial negatives.

Ping-style APD. A plain APD burst discards a candidate on the first timeout; we observed that transient loss ($1-3\%$) inflates false negatives to $\sim 7\%$. Our variant issues up to $r_{\max}$=5 retries per sub-address spaced by 200 ms and declares a positive as soon as all 16 sub-addresses reply, cutting the false-negative rate to $\sim$0.6% at $\leq 1.2\times$ probe overhead in the worst case.

Online fine-tuning. Positives grow $\mathcal{F}$ and a buffer $\mathcal{P}$; negatives grow $\mathcal{N}$. Whenever the cumulative probe count crosses a snapshot in $\{0.5, 2.5, 4.5, 6.5\}$ M, we take one gradient step on a balanced batch of 16 samples from $\mathcal{P} \cup \mathcal{N}$ with learning rate 10^{-5} (two orders below pre-training) and gradient-norm clip 1.0. This chunked back-propagation fits the fine-tune step in a 24 GB RTX 4090 (a naive full-batch update triggered OOM at $\sim$17 GB).

Algorithm 1 summarises the loop: L1/L2 enforce the in-range exclusion of Eq. (2); FINETUNE realises the feedback that closes the control loop.

Algorithm 1 Online FRP discovery with in-range-aware feedback

Require: pretrained θ_0, seed set $\mathcal{S}$, budget B

$\mathcal{F} \leftarrow \mathcal{S}$, $b \leftarrow 0$
while $b < B$ **do**
  $\mathcal{C} \leftarrow$ DS-HMAMBA$(\theta; \text{ctx})$ ▷ generate candidates
  $\mathcal{C}_1 \leftarrow \{c \in \mathcal{C} \mid c \notin \mathcal{F}\}$ ▷ L1 filter
  $\mathcal{C}_2 \leftarrow \{c \in \mathcal{C}_1 \mid \nexists s \in \mathcal{F},\ c \prec s\}$ ▷ L2 filter
  for all $c \in \mathcal{C}_2$ **do**
    $r \leftarrow$ APD-PING$(c, 16)$
    $b \leftarrow b + 16$
    if $r =$ FRP **then**
      $\mathcal{F} \leftarrow \mathcal{F} \cup \{c\}$
    end if
  end for
  FINETUNE$(\theta; \mathcal{C}_2, r)$
end while
return $\mathcal{F}$

V. REDUNDANCY ANALYSIS OF EXISTING GENERATORS

We now quantify limitation **L1** identified in §I. For every FRP c confirmed by an algorithm we test whether $c \preceq s$ for some $s \in \mathcal{S}_0$; if so we count it as *in-range* and remove it from

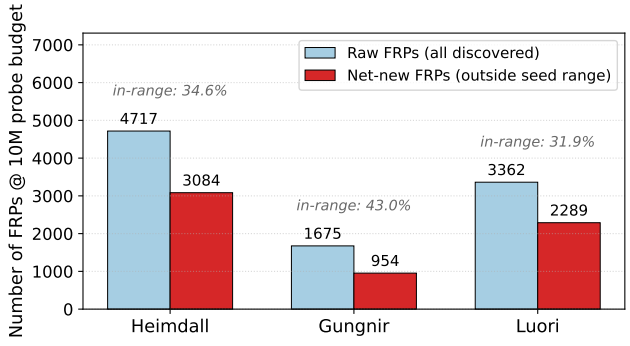


Fig. 3: Raw vs. net-new FRPs at the 10M-packet budget. Darker bars are FRPs whose prefixes are *not* covered by the aligned seed set $\mathcal{S}_0$. Although Heimdall still shares a 34.6% overlap with $\mathcal{S}_0$ (due to seed lists released after its training cut-off), it delivers 3,084 net-new FRPs against Gungnir's 954 — a 3.23× improvement, and 1.35× that of Luori.

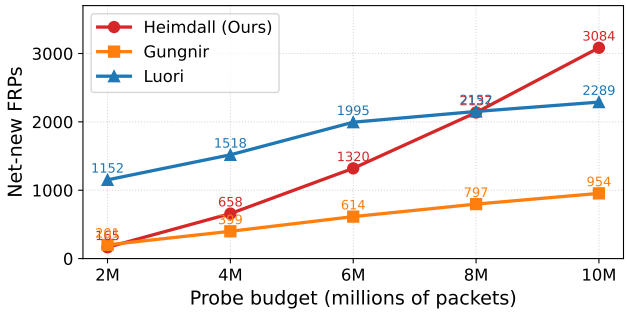


Fig. 4: Net-new FRPs as a function of the probing budget. Gungnir grows almost linearly from a very low base; Luori saturates around 8M; Heimdall's in-range-aware feedback loop produces a super-linear trajectory that overtakes both baselines by $B \geq 10$M.

the net-new pool. We sweep $B \in \{2, 4, 6, 8, 10\}$ M packets and repeat the analysis for all three generators under identical seeds and alignment.

Redundancy at the target budget. At 10M, 43.0% of Gungnir's confirmed FRPs are in-range (954 net-new out of 1,675); Luori's fraction is 31.9% (2,289/3,362) and Heimdall's is 34.6% (3,084/4,717)—in absolute terms Heimdall delivers 3.23× more net-new FRPs than Gungnir and 1.35× more than Luori (Table I(a), Fig. 3). Heimdall's residual overlap reflects seed heterogeneity: the alignment pipeline behind $\mathcal{S}_0$ ingests APD-derived aggregations and third-party releases published after our training cut-off, sources the L2 filter cannot cover; against the training seed itself the L2 filter guarantees 0% in-range candidates.

How net-new grows with the budget. The distinctive property of Heimdall is its *trajectory* (Table I(b), Fig. 4). Gungnir grows nearly linearly from a low base (201 → 954) while Luori saturates once its seed-driven expansion is exhausted (1,152 → 2,289, essentially flat between 8M and 10M). Heimdall starts *below* Gungnir at 2M (165 vs. 201), matches Luori at 8M (2,137 vs. 2,152) and by 10M is 3.23× Gungnir and 1.35× Luori. This super-linear regime is exactly what the in-range-aware feedback loop is designed to produce.

VI. Evaluation

A. Experimental Setting

Dataset. 227,894 IPv6 BGP prefixes from RouteViews [21], following [13]; Luori's public FRP set [12] as the initial seed $\mathcal{S}_0$ (182,942 ICMPv6 FRPs over 4,643 routing prefixes). *Hyper-parameters.* DS-HMamba: 6 layers, hidden dim d=128, state dim N=16, Conv-1D width 4; pre-training runs 50 epochs with Adam (lr=10^{-3}, batch 512). Online fine-tuning: batch 16 with one gradient step per snapshot. APD: 16 probes/prefix, max 5 retries, 2 s timeout. *Baselines.* Luori [12] (MCTS-RL) and Gungnir [13] (Transformer AR), run with authors' released code and identical hyper-parameters except probing budget. *Budget grid.* $B_{\text{total}} \in \{2, 4, 6, 8, 10\}$ M packets. *Metrics.* $|\mathcal{F}|$, CovSpace $= \sum_{f \in \mathcal{F}} 2^{128-\ell(f)}$, CovAS, CovRouting, CovCountry, CovCat.

B. Overall Performance

Table II reports the *raw* performance grid and Table III the same experiment restricted to FRPs not covered by $\mathcal{S}_0$ (*net-new*). Fig. 5 visualises the net-new view. Key findings. *Raw FRPs.* Heimdall dominates uniformly, with a 2.8–6.5× advantage over Gungnir and 1.3−1.4× over Luori (Table II). *Net-new FRPs.* Once seed overlap is removed the picture becomes more dramatic: Gungnir plateaus below 10^3 (201 → 954), Luori saturates at ∼2,300, only Heimdall shows super-linear growth (165 → 3,084, i.e. 18.7× from 2M to 10M vs. 4.7× for Gungnir and 2.0× for Luori); at 10M Heimdall contributes 3,084 genuinely new FRPs (3.23× Gungnir, 1.35× Luori). *Address space.* Heimdall covers 2.04×10^{32} addresses at 10M vs. 3.46×10^{31} for Gungnir (5.9× raw); the net-new gap widens to ×20 (7.5×10^{31} vs. 3.7×10^{30}). *Topological breadth.* At 10M raw, Heimdall touches 314 ASes and 44 countries vs. Gungnir's 97/25 and Luori's 7/2; among net-new, 230/41 vs. 49/19 (Gungnir) and 4/2 (Luori). *Redundancy trajectory.* Heimdall's in-range fraction drops from 93% at 2M to 35% at 10M as the feedback loop kicks in, whereas Gungnir stays at ∼43% and Luori at ∼30% throughout.

C. Ablation Studies

Table IV(a) disables one component at a time and reports both raw $|\mathcal{F}|$ and net-new NN. *Inner scan.* Disabling it drops net-new from 1,320 to 802 (−39%), confirming the intra-AS signal identified in §III-C. *L2 SeedFilter.* Without the sub-prefix filter raw goes up to 4,015 (the model happily re-confirms seeds) while net-new plummets to 472 (−64%)—in-range candidates actively displace productive probes. *Online fine-tuning.* Freezing θ saturates the generator at 951 net-new FRPs (close to Luori's plateau); the loop is responsible for the super-linear regime of Fig. 5(a). *AS gate.* Uniform averaging over the two scans costs −16% raw and −21% net-new. *Causal LM vs. MLM.* A left-to-right causal loss underperforms MLM by −12%/ − 18% (raw/net-new), because it forces fine bits to attend only to preceding bits. Regarding sampling temperature, p=0.6 maximises CovAS/CovRouting; p=0.4 over-exploits and inflates redundancy above 60%.

D. Resource Consumption

Table IV(b) compares pre-training, per-candidate inference and peak GPU memory on a single RTX 4090 (24 GB HBM, PyTorch 2.3, CUDA 12.4). DS-HMamba trains in 6.8 h (matching Gungnir); inference is 1.7× faster per candidate than Gungnir thanks to the $O(L)$ outer scan; peak GPU memory is 3.2 GB, well within a commodity GPU (naive full-batch fine-tuning hit OOM at ∼17 GB, motivating the chunked back-propagation of §IV-E).

VII. Discussion and Conclusion

Why in-range-aware feedback wins. Every probe issued by Heimdall is guaranteed to fall outside $\mathcal{S}_0$: a confirmed FRP directly enlarges $\mathcal{S}_0$, a negative refines the generator away from that region. In contrast, Gungnir's top-p decoder concentrates on high-density training regions that coincide

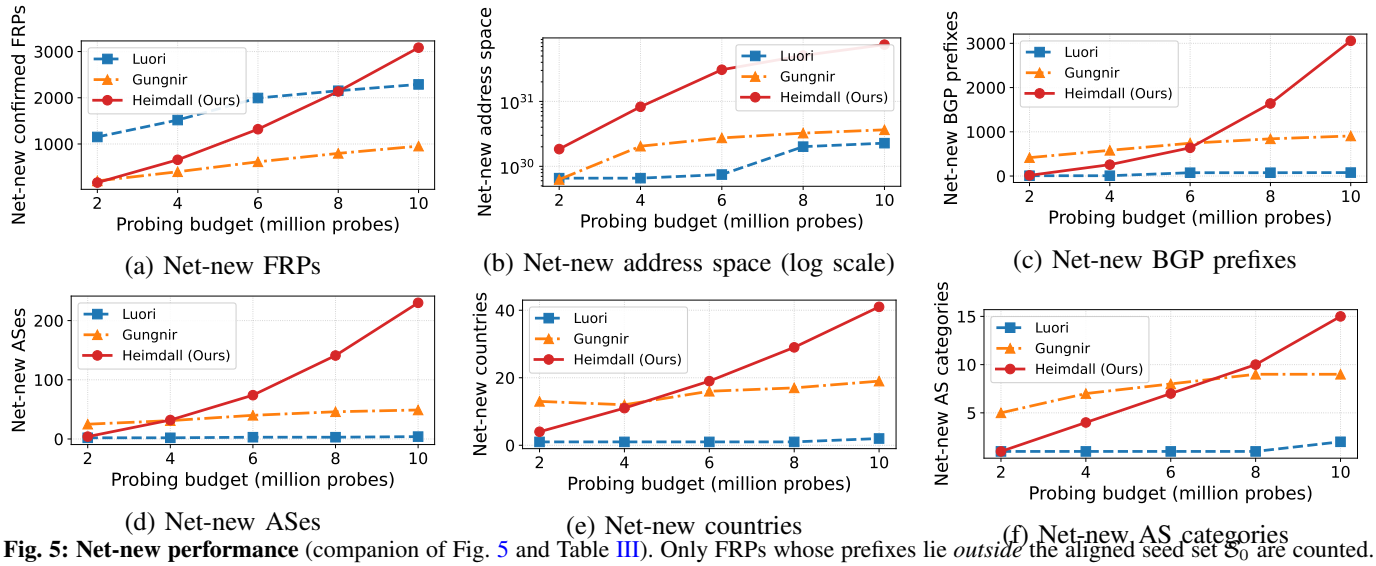
TABLE I: Redundancy of the three generators against the aligned public seed set $\mathcal{S}_0$. *Raw* counts confirmed FRPs; *In-range* counts those covered by $\mathcal{S}_0$; *Net-new* = *Raw* − *In-range*. (a) snapshot at the target 10M budget. (b) net-new FRPs as a function of the probing budget; Heimdall's growth is super-linear while Gungnir and Luori saturate. Data collected on the RouteViews view [21] on the same day.

(a) 10M snapshot

Method	Raw	In-range	**Net-new**
Gungnir [13]	1,675	43.04%	954
Luori [12]	3,362	31.92%	2,289
Heimdall (Ours)	**4,717**	**34.62%**	**3,084**

(b) Net-new FRPs across budgets

Method	2M	4M	6M	8M	10M
Gungnir	201	399	614	797	954
Luori	1,152	1,518	1,995	2,152	2,289
Heimdall (Ours)	165	**658**	**1,320**	**2,137**	**3,084**
Heimdall/Gungnir	0.82×	1.65×	2.15×	2.68×	**3.23×**
Heimdall/Luori	0.14×	0.43×	0.66×	0.99×	**1.35×**



(a) Net-new FRPs (b) Net-new address space (log scale) (c) Net-new BGP prefixes

(d) Net-new ASes (e) Net-new countries (f) Net-new AS categories

Fig. 5: Net-new performance (companion of Fig. 5 and Table III). Only FRPs whose prefixes lie *outside* the aligned seed set $\mathcal{S}_0$ are counted. Heimdall is the only method whose net-new curve grows super-linearly with the budget and overtakes both baselines from 6M onwards.

with the seed set; this is why its redundancy stays flat at ∼43% regardless of the budget, while Heimdall's drops from 93% at 2M to 35% at 10M.

Limitations and generality. The seed trie captures only $\mathcal{S}_0$ available at start-up; FRPs released *after* the training cut-off are still probed as fresh candidates, producing the residual 34.6% overlap observed at 10M (§V). A periodic seed refresh against a live community FRP mirror would close this gap. DS-HMamba is pre-trained on ICMPv6 only; TCP/UDP extensions require re-training the scans but leave the feedback loop unchanged (context tokens are protocol-agnostic).

Ethics. We follow the Menlo Report [22] and community best practices [23], [25], [37]: rate-limited scanning ($\leq$ 1 Mbps), a single measurement host, an opt-out mailing list, and no penetration testing on any identified host.

Conclusion. We show that state-of-the-art FRP generators waste 32%–43% of their probing budget on candidates already contained in the aligned public seed range. **Heimdall** combines a dual-scan hierarchical selective-state-space generator with an in-range-aware online feedback loop; under identical budgets it discovers 2.8× more raw and 3.23× more *net-new* FRPs than Gungnir, covers 5.9× more address space and 3.2× more ASes, while remaining 1.7× faster to sample. Code, data and the enlarged FRP list will be released at https://anonymous.4open.science/r/Heimdall. Future work will explore periodic seed refresh and multi-protocol extension to TCP/UDP.

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TABLE II: Raw performance. All FRPs *discovered* by each generator under identical probing budgets on the public IPv6 routing view; no seed-range filter is applied. SPACE numbers are in addresses.

	Luori [12]					**Gungnir** [13]					**Heimdall (Ours)**				
Budget	2M	4M	6M	8M	10M	2M	4M	6M	8M	10M	2M	4M	6M	8M	10M
$\|\mathcal{F}\|$	1,889	2,255	2,820	2,994	3,362	376	730	1,089	1,380	1,675	**2,449**	**3,076**	**3,662**	**4,213**	**4,717**
CovSpace	4.0e30	4.0e30	4.4e30	6.0e30	7.8e30	1.0e31	1.6e31	2.1e31	2.8e31	3.5e31	**1.5e32**	**1.6e32**	**1.6e32**	**1.8e32**	**2.0e32**
CovRouting	8	8	78	80	83	566	879	1,138	1,296	1,433	**1,634**	**1,745**	**2,120**	**3,378**	**4,744**
CovAS	2	2	4	6	7	44	56	70	81	97	**143**	**155**	**195**	**254**	**314**
CovCountry	1	1	1	2	2	20	21	23	24	25	**36**	**35**	**37**	**40**	**44**
CovCat	1	1	1	2	2	6	9	9	9	9	**9**	**11**	**12**	**13**	**15**

TABLE III: Net-new performance. Same experiment as Table II, but each metric only counts FRPs whose prefixes are *not* covered by the aligned public seed set $\mathcal{S}_0$ (i.e., truly new discoveries). Numbers in parentheses are the *in-range redundancy* $1-(\text{net-new}/\text{raw})$ of $|\mathcal{F}|$: lower is better in relative terms, but the absolute net-new count is what ultimately matters for a hitlist. Heimdall's net-new $|\mathcal{F}|$ grows super-linearly with the budget, ending at $3.23\times$ Gungnir and $1.35\times$ Luori.

	Luori [12]					**Gungnir** [13]					**Heimdall (Ours)**				
Budget	2M	4M	6M	8M	10M	2M	4M	6M	8M	10M	2M	4M	6M	8M	10M
$\|\mathcal{F}\|$	1,152	1,518	1,995	2,152	2,289	201	399	614	797	954	165	**658**	**1,320**	**2,137**	**3,084**
(redund.)	39%	33%	29%	28%	32%	47%	45%	44%	42%	43%	93%	79%	64%	49%	35%
CovSpace	6.6e29	6.6e29	7.5e29	2.0e30	2.3e30	6.3e29	2.0e30	2.7e30	3.2e30	3.7e30	**1.8e30**	**8.3e30**	**3.1e31**	**5.1e31**	**7.5e31**
CovRouting	7	7	75	75	78	**416**	**581**	**742**	841	906	15	257	638	**1,640**	**3,057**
CovAS	2	2	3	3	4	**25**	31	40	46	49	4	**32**	**74**	**141**	**230**
CovCountry	1	1	1	1	2	**13**	**12**	16	17	19	4	11	**19**	**29**	**41**
CovCat	1	1	1	1	2	**5**	**7**	**8**	9	9	1	4	7	**10**	**15**

TABLE IV: Ablation study (left, at $B_{\text{total}} = 6\text{M}$; net-new measured against $\mathcal{S}_0$) and **resource consumption** (right; NVIDIA RTX 4090, PyTorch 2.3, CUDA 12.4).

Configuration	$\|\mathcal{F}\|$	NN	CovAS	Notes
Full Heimdall	**3,662**	**1,320**	**195**	—
– w/o Inner Scan	2,528	802	141	-31% raw
– w/o L2 SeedFilter	4,015	472	158	-64% net-new
– w/o Online Fine-Tuning	2,954	951	158	plateau at snap-1
– w/o AS Gate (avg pool)	3,092	1,041	173	-16% raw
– causal LM (no MLM)	3,218	1,087	178	-12% raw

(a) Ablation study

Generator	Pre-tr. (h)	Infer. (ms/cand.)	Peak GPU (GB)
Luori [12]	5.6	4.1	2.4
Gungnir [13]	7.1	2.4	5.8
Heimdall (Ours)	**6.8**	**1.4**	**3.2**

(b) Resource consumption